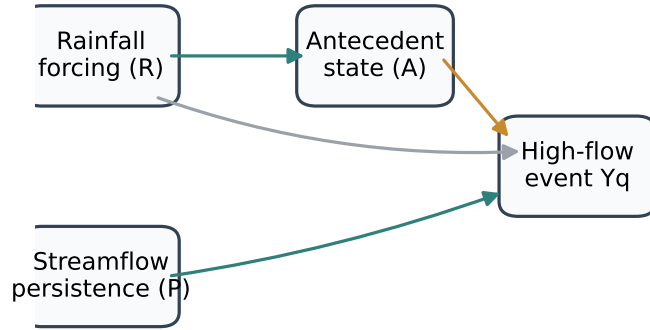


a Information partition tested in this study



Tested increment: $S(Y_q; R, P, A) - S(Y_q; R, P)$

State value is retained only when it survives rainfall plus streamflow persistence.

b Competing interpretations

H1: memory partition

A adds value where P has not absorbed catchment memory.

H2: storage ceiling

Storage-related regimes organize where P and A overlap.

H0: benchmark null

Rainfall plus persistence is sufficient for the admitted target.

c Evidence retained as support or boundary

Event definition	supports boundary
Persistence boundary	supports boundary
Storage moderation	supports boundary
Temporal recurrence	supports boundary
Spatial portability	limits portability
Era-region portability	limits portability

d Inference boundary

Not a global phase transition

Event sensitivity is a threshold-surface result.

Not globally portable

Temporal recurrence and spatial transfer answer different questions.

Not a closed storage mechanism

Storage proxies moderate heterogeneity, not causal closure.

Observed anchors not authority

Observed supports challenge directionality only.